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Hierarchical Conditional-Probability-Based Adaptive Patch Sampling

Organ-Wise Learning States and Error Types for 3D Abdominal CT Multi-Organ Segmentation

Keywords Medical Image Segmentation · Adaptive Patch Sampling · Hierarchical Conditional Probability · nnU-Net · 3D CT

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Related Research

Existing sampling strategies for 3D medical image segmentation

1 Standard Patch Sampling

nnU-Net — Isensee et al. (2021) [2]

- Self-configuring biomedical segmentation framework
- Random patch extraction and foreground oversampling
- No explicit organ-wise learning-state adaptation

2 Adaptive Patch Sampling

Park et al. (2026) [3]

- Adaptive sampling based on prediction errors and uncertainty

He et al. (2022) [4]

- Online hard-patch mining with shape models and bandits

Our Research Focus:

Organ state → Error type → Patch center

Hierarchical sampling based on organ-wise learning states and semantic error types.

[2] Isensee et al., “nnU-Net,” Nature Methods, 18, 203–211, 2021.
[3] Park et al., “Adaptive Patch Sampling and Location-Aware Reasoning,” Scientific Reports, 16, 22461, 2026.
[4] He et al., “Online Hard Patch Mining,” IEEE J. Biomed. Health Inform., 26(6), 2648–2659, 2022.

Research Objectives & Contributions

Four complementary components of the proposed sampling policy

1 Organ-Wise Learning States

- Estimate organ difficulty from Dice deficits
- Stabilize learning states with EMA
- Prioritize underperforming organs

2 Error-Type Awareness

- Interior miss · Boundary disagreement
- Exterior false positive
- Maintain error-type candidate pools

3 Hierarchical Conditional Sampling

- Select organ k → error type r → center c
- Use conditional selection probabilities
- Extract a 3D patch around the selected center

4 Controlled Ablation Study

- B0: foreground oversampling · B1: error candidates
- A1: organ adaptation · P: full hierarchy
- Compare under matched training conditions

Research Objective Assess 3D segmentation learning at a fixed update budget without changing nnU-Net.

Proposed Sampling Framework

Training-only state observation → conditional selection → 3D patch extraction

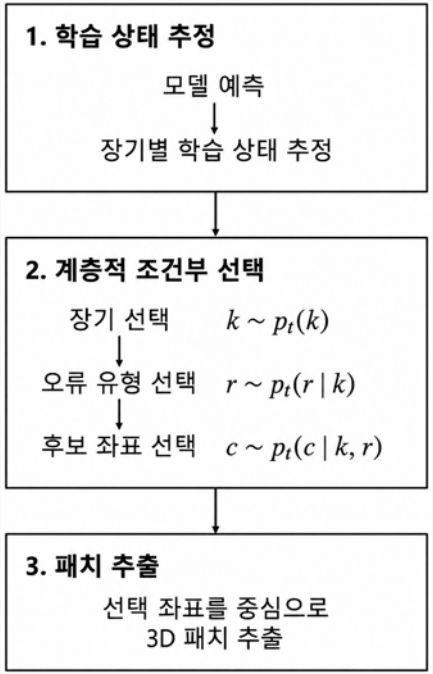


Fig. 1 · Workflow from the manuscript

1 **Observe learning state**

Use prediction–label agreement on training observer patches to estimate organ difficulty.

2 **Choose organ, error type, and center**

Select k, then r conditioned on k; draw c uniformly from the eligible candidate pool.

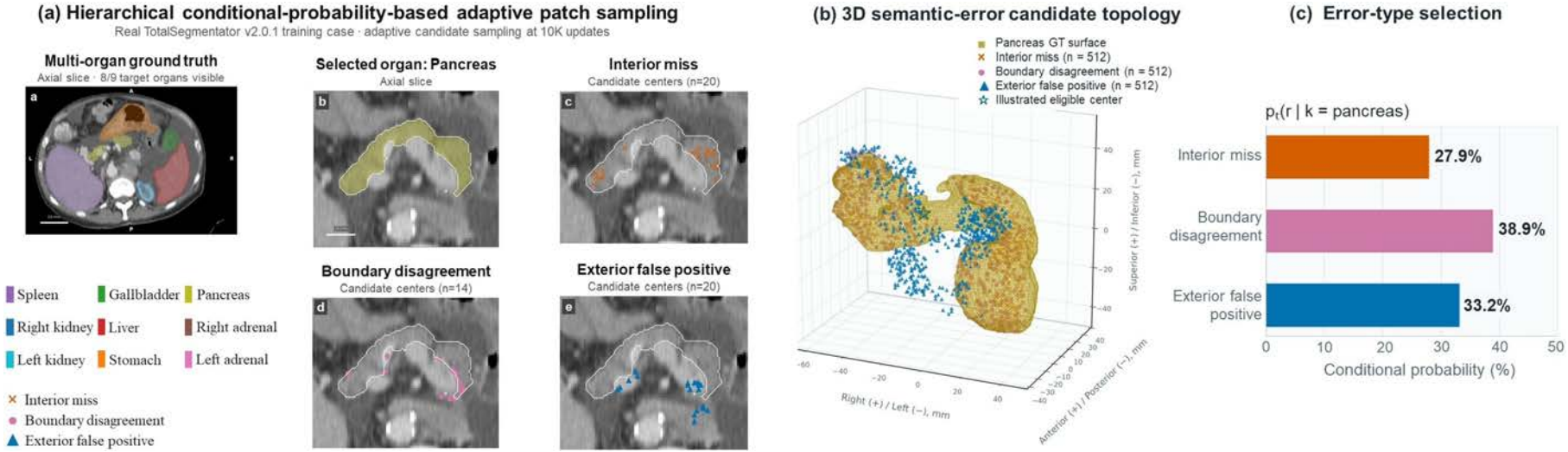
3 **Extract a 3D patch**

Crop the fixed-size patch around the selected (z, y, x) center for model training.

Guided-branch joint selection probability

$$p_t(k, r, c) = p_t(k)\, p_t(r \mid k)\, p_t(c \mid k, r)$$

Hierarchical Sampling Mechanism



Note: candidate markers are bounded reservoirs; the highlighted center is illustrative, not a logged random draw.

Organ: pancreas	Error: boundary	Candidate pool	One center path
p(k) = 0.1340	p(r | k) = 0.3889	512 centers	p(k, r, c) ≈ 0.0001018

Experimental Setup & Ablation

Same architecture and training budget; only the patch sampling policy is varied

Dataset and training configuration

Dataset	TotalSegmentator v2.0.1 [1]
Eligible cases	602 · nine abdominal organs
Train / Val / Test	525 / 28 / 49
Backbone	nnU-Net v2.8.1 · 3d_fullres [2]
Spacing / patch	1.5 mm · 160 × 112 × 128
Batch / budget	2 · 30,000 optimizer updates

Sampling-policy ablation

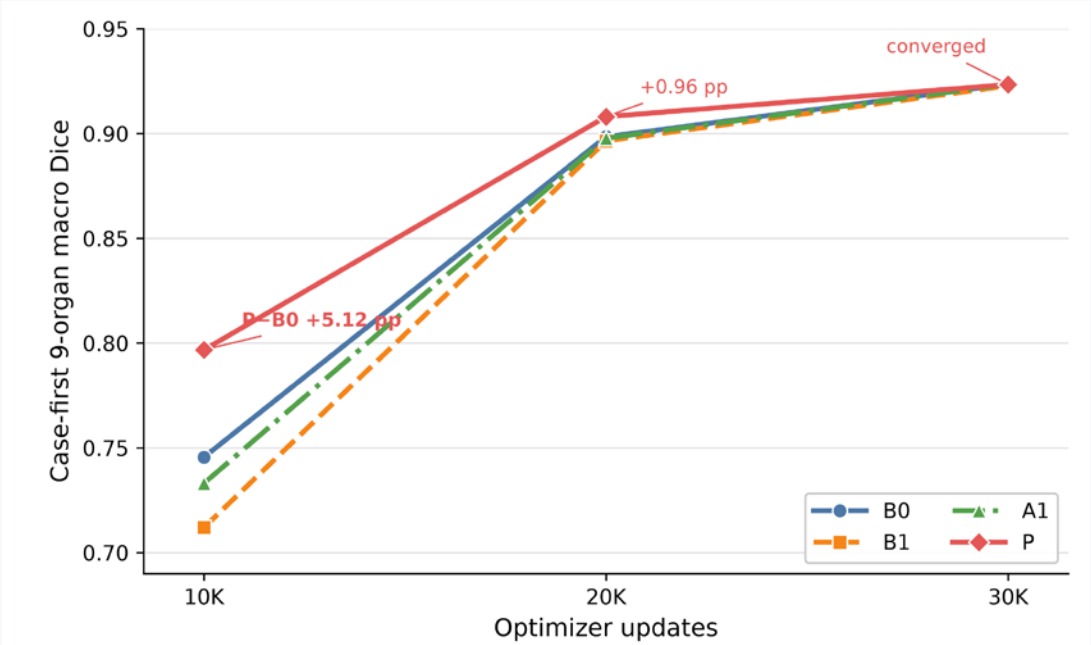
Policy	Candidates	Organ	Error
B0	—	—	—
B1	✓	—	—
A1	✓	✓	—
P	✓	✓	✓

Main metric: $$S_m(u) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{9}\sum_{k=1}^{9}\mathrm{Dice}_{i,k}^{(m,u)}\right)$$

[1] Wasserthal et al., Radiol. Artif. Intell. (2023). [2] Isensee et al., Nature Methods (2021).

Validation Results

Single-run discovery on 28 validation cases



P versus B0

10K **+5.1228 pp**

B0 0.7455 · P 0.7968

20K **+0.9603 pp**

B0 0.8984 · P 0.9080

30K **−0.0139 pp**

B0 0.9236 · P 0.9234

Early P–B0 gap decreased with training; 30K macro Dice was nearly identical.

Conclusions & Future Work

A hierarchical sampling-policy design for 3D abdominal CT multi-organ segmentation

Conclusions

- Introduced organ-aware and error-type-aware conditional patch allocation.
- Maintained the nnU-Net architecture while changing the sampling policy.
- Observed +5.12 pp Dice at 10K in the representative run.
- The difference diminished and was nearly zero by 30K.

Future Work

01 Robust evaluation

Assess stability across training conditions and external datasets.

02 Allocation refinement

Explore more stable organ-state and error-type weighting.

03 Computational efficiency

Measure observer updates, overhead, and time-to-accuracy.